

DECK 16 · WEEK 8 · MORNING 1

Fixed Income IV

Credit & Securitization

How credit risk is measured, analysed across issuer types, and repackaged into asset-backed securities

CFA Level I · Vol 6 Learning Modules 14–19

Roadmap — Fixed Income IV — Credit Risk, Credit Analysis & Securitization

5 sections · ~2 hours · Built bottom-up from CFA readings

01

Credit-risk framework

POD · LGD · EAD · EL = $POD \times LGD$ · recovery rates · rating scales · spread decomposition · spread and price

02

Sovereign credit analysis

Ability AND willingness · sovereign immunity · five qualitative factors · fiscal, growth, external ratios

03

Corporate credit analysis

Eight Cs (5 bottom-up + 3 top-down) · margin, coverage, leverage · seniority · notching · subordination

04

Securitization structures

SPE · true sale · credit tranching · three internal enhancements · non-mortgage ABS · CLOs · covered bonds

05

MBS specifics

RMBS vs CMBS · agency vs non-agency · WAC / WAM · contraction vs extension · CMOs · PAC · balloon risk

Learning objectives

By the end of this deck you should be able to:

1. Decompose credit risk into probability of default, loss given default and expected exposure, and compute expected loss in currency or percentage terms.
2. Interpret the rating scales of Moody's, S&P and Fitch, the Baa3 / BBB- line, credit migration risk, and decompose a yield into benchmark, liquidity and credit spread.
3. Evaluate a sovereign on ability AND willingness to pay, using the five qualitative factors and the fiscal, growth and external ratio families.
4. Apply the eight Cs of credit analysis, compute the four ratios of Exhibit 3, and explain notching and structural subordination.
5. Rank claims in a default waterfall, apply recovery rates by seniority, and describe securitization mechanics — SPE, true sale, credit tranching, the three internal enhancements.
6. Analyse MBS using WAC, WAM and the pass-through rate, and distinguish contraction from extension risk, sequential-pay CMOs, PAC tranches and CMBS balloon risk.

SECTION 01

Credit-risk framework ($\text{POD} \times \text{LGD} = \text{EL}$)

Credit risk is the expected economic loss from a borrower failing to pay in full and on time — and it decomposes into a probability and a severity

POD is a probability, typically annualised. LGD is a currency (or percentage) amount: how big the claim is, times how much of it is lost.

$$EL = POD \times LGD$$

$$LGD = EE \times (1 - RR)$$

where: POD = probability of default; EE = expected exposure (= EAD); RR = recovery rate; $(1 - RR)$ = loss severity. LGD may be stated in currency (EUR 450,000) or as a percentage of principal (45%) — the percentage form is what allows comparison across borrowers.

1 Probability of default (POD)

Conditional, forward-looking likelihood the issuer fails to make full and timely payment, assuming no prior default. Typically measured over a 12-month horizon. Driven by profitability, coverage and leverage.

2 Expected exposure (EE = EAD)

The size of the investor's claim at the moment of default: face value plus accrued interest, less the current market value of available collateral.

3 Recovery rate (RR) → LGD

RR is the percentage of the claim recovered; $(1 - RR)$ is loss severity. $LGD = EE \times (1 - RR)$. LGD is a function of seniority — secured debt recovers more, junior subordinated debt often has $LGD \approx 100\%$.

Source: CFA Institute, Curriculum Vol 6, Learning Module 14, Exhibit 4 and Equations 1–2, printed pp. 349–351.

Expected loss is probability times severity times claim size — and the fair credit spread is an approximation of that same product

Collateral is netted off expected exposure BEFORE the recovery haircut is applied. Miss that step and you overstate the loss by a third.

$$EL = POD \times (EE - \text{Collateral}) \times (1 - RR)$$

where: And Credit spread $\approx POD \times LGD$: the investor is fairly compensated when the spread equals the annualised expected loss. Curriculum Example 2 — BRWA has POD 1% and LGD 80% $\Rightarrow EL$ 0.80% against a 90 bps G-spread, so more than fairly compensated.

CFA LM 14 SELF-ASSESSMENT Q2 — EUR 500,000 LOAN (printed pp. 342-343)

Probability of default 5%. Expected exposure EUR 400,000. Collateral EUR 100,000. Recovery rate 90%.

1. Net the collateral off exposure: $400,000 - 100,000 = \text{EUR } 300,000$
2. Loss severity: $1 - RR = 1 - 0.90 = 10\%$
3. LGD in currency: $300,000 \times 10\% = \text{EUR } 30,000$
4. $EL = POD \times LGD$: $0.05 \times 30,000 = \text{EUR } 1,500$

$EL = \text{EUR } 1,500 = 0.30\%$ of the EUR 500,000 face. A spread above 30 bps more than compensates.

Source: CFA Institute, Curriculum Vol 6, Learning Module 14, Self-Assessment Q2 (printed pp. 342–343) and Example 2 (printed pp. 351–352).

The three agencies use parallel symbol scales, and the market splits the universe at Baa3 / BBB– — below that line, bonds are speculative grade

Ratings are commissioned and paid for by the issuer, and they measure default risk — not price. They are an opinion on expected loss, not a buy signal.

Rating grade	Moody's	S&P / Fitch	Grade	Description
High-Quality Grade	Aaa	AAA	IG	Highest quality, lowest risk
High-Quality Grade	Aa1 – Aa3	AA+ – AA–	IG	Very high quality, very low risk
Upper-Medium Grade	A1 – A3	A+ – A–	IG	High quality, low credit risk
Low-Medium Grade	Baa1 – Baa3	BBB+ – BBB–	IG	Good quality, moderate risk
Low Grade / Speculative	Ba1 – Ba3	BB+ – BB–	HY	Speculative, substantial risk
Low Grade / Speculative	B1 – B3	B+ – B–	HY	Highly speculative, high risk
Low Grade / Speculative	Caa1 – Caa3	CCC+ – CCC–	HY	Default a real possibility
Low Grade / Speculative	Ca	CC	HY	Default occurring or imminent
Default	C	D	D	In default, little recovery

THE B a a 3 / B B B – LINE, AND MIGRATION RISK

Credit migration (downgrade) risk is the risk that creditworthiness deteriorates — no default required. Ratings also trigger covenants and step-up coupons. IG issuers borrow more consistently and more cheaply than HY issuers.

Source: CFA Institute, Curriculum Vol 6, Learning Module 14, Exhibit 7, printed pp. 356–357. S&P and Fitch are identical at every notch shown; D means already in default, while Moody's C-rated bonds are likely, but not necessarily, in default.

Four ratio families drive POD – profitability, coverage and two measures of leverage – while LGD is set separately by seniority

Curriculum wording: higher profitability and higher coverage, coupled with lower leverage, are associated with a lower POD and higher credit quality.

Driver	Ratio in Exhibit 5	Exhibit 5 axis range	As rating improves
Profitability	EBIT margin	0% – 30%	Rises
Coverage	EBIT / Interest expense	0x – 20x	Rises
Leverage (earnings)	Debt / EBITDA	0x – 8x	Falls
Leverage (cash flow)	Cash flow / Net debt	0% – 100%	Rises

WHAT EXHIBIT 5 ACTUALLY SHOWS – AND WHAT IT DOES NOT

It plots each ratio against rating category from Aaa down to Ca, establishing direction and rank order. It does NOT publish a threshold table. Ratio levels for a given rating vary widely across industries and over time, so any single cut-off is an industry judgement, not a rule.

POD SIDE

Set by the issuer's own financials: stable profits, enough cash flow to cover debt service, limited reliance on debt.

LGD SIDE (EXHIBIT 6)

Set by the nature and seniority of the claim: secured debt = lower LGD, unsecured debt = higher LGD. Ratios do not move it.

¹ Exhibit 5 is four scatter charts of ratio against rating category. It shows ordering and axis ranges only — no numeric threshold per rating.

Source: CFA Institute, Curriculum Vol 6, Learning Module 14, Exhibits 5 and 6, printed pp. 352–354.

A corporate yield is a benchmark plus a liquidity spread plus a credit spread — and a change in that spread hits price through duration and convexity

The same duration and convexity statistics apply whether the yield change comes from the benchmark or from the spread. Spread narrowing raises the holding-period return.

$$\% \Delta PV_{full} = - (\text{AnnModDur} \times \Delta S) + \frac{1}{2} \times \text{AnnConvexity} \times (\Delta S)^2$$

where: ΔS = the change in spread. Scale convexity to the same order of magnitude as duration squared: a reported convexity of 0.23 is used as 23. Spreads are narrowest at or near the top of the credit cycle and widest at or near the bottom.

DECOMPOSING THE SPREAD — ROMANIA 2049 EUROBOND

Mid price 124.00 on a 4.625% bond with 28 years left $\Rightarrow$ yield 3.2988%.

Benchmark: German bund 0.2350% (23.50 bps).

Yield spread: 3.2988% – 0.2350% = 306.38 bps.

Liquidity spread from the bid/offer: yield at the bid 3.3588% less yield at the offer 3.2396% = 11.92 bps.

Credit spread = 306.38 – 11.92 = 294.46 bps.

LM 14 SELF-ASSESSMENT Q6 — HY BOND, SPREADS +100 bps

Modified duration 4.5; reported convexity 0.23, rescaled to 23; $\Delta \text{Spread} = +1.00\%$.

1. Duration: $-4.5 \times 1.00\% = -4.500\%$
2. Convexity: $\frac{1}{2} \times 23 \times (1.00\%)^2 = +0.115\%$
3. Sum: $-4.500\% + 0.115\% = -4.385\%$

-4.385%. Convexity softens the blow by 11.5 bps.

SECTION 02

Sovereign and non-sovereign government issuers

Sovereign creditworthiness rests on ability to pay AND willingness to pay — because sovereign immunity blocks investors from forcing bankruptcy

Five qualitative factors, with Government Institutions and Policy sitting above the other four. Willingness is the factor with no ratio behind it.

GOVERNMENT INSTITUTIONS

Rule of law, property rights, political stability, data transparency — and the willingness-to-pay record.

FISCAL FLEXIBILITY

Ability to adjust revenue and spending; fiscal discipline over the cycle; prudent use of debt.

MONETARY EFFECTIVENESS

Policy credibility, exchange-rate regime, central-bank independence, depth of the domestic debt market.

ECONOMIC FLEXIBILITY

Size, per-capita income, diversification, competitiveness, adaptability to shocks.

EXTERNAL STATUS

Global currency status, access to external funding, geopolitical risk.

SOVEREIGN IMMUNITY IN PRACTICE — URUGUAY, MAY 2003

After a banking crisis and a forced float of the peso in June 2002, Uruguay agreed an IMF-supported debt exchange extending bond maturities by five years, with 90% of bondholders consenting. Although the transaction was voluntary, analysts deemed it a default in substance: holders were not adequately compensated for the delay, and non-participating holdouts were subordinated.

Source: CFA Institute, Curriculum Vol 6, Learning Module 15, Exhibits 1–2 and Example 1, printed pp. 383–386.

Quantitative sovereign analysis uses three ratio families – fiscal strength, economic growth and stability, and external stability

The numerator is debt or interest, as for a company. The denominator is GDP (what can be taxed), government revenue (what is actually collected), or FX reserves (external firepower).

Family (Exhibit 3)	Ratio	Definition	Higher ratio ⇒
Fiscal — debt burden	Debt / GDP	General govt debt ÷ GDP	Weaker credit
Fiscal — debt burden	Debt / Revenue	General govt debt ÷ revenue	Weaker credit
Fiscal — affordability	Interest / GDP	Interest payments ÷ GDP	Weaker credit
Fiscal — affordability	Interest / Revenue	Interest payments ÷ revenue	Weaker credit
Growth and stability	Real GDP growth	Average growth, and its std dev	Stronger credit
Growth and stability	Per capita GDP	GDP ÷ population	Stronger credit
External stability	FX reserves / GDP	FX reserves ÷ GDP	Stronger credit
External stability	Reserve ratio	FX reserves ÷ external debt	Stronger credit

THE DATA PROBLEM – AND WHY THERE IS NO THRESHOLD TABLE

Sovereigns do not publish IFRS accounts. Analysts rely on government statistics that vary in quality and timing, lack cross-country comparability and are revised. Exhibit 5 plots Debt/Revenue and Interest/GDP against Moody's categories, but publishes no bands: the metrics are weighted and combined, never read in isolation.

Source: CFA Institute, Curriculum Vol 6, Learning Module 15, Exhibits 3, 4, 6 and 7, printed pp. 387–392.

Ratios rank the peers but do not decide the rating — Thailand has the strongest reserves of the five and is still rated below Malaysia

Malaysia wins on competitiveness and lower reliance on foreign-currency debt. The Philippines matches Thailand because remittances — nearly 10% of GDP — are a foreign-currency inflow.

Sovereign	Moody's FC rating	Reserve ratio	FX reserves / GDP	What Example 6 says
Malaysia	A3	54%	11.9%	Highest per-capita GDP
Thailand	Baa1	102%	16.4%	Best reserves, lower rating
Philippines	Same as Thailand	96%	10.3%	Remittances ~10% of GDP
Vietnam	Ba3	84%	9.9%	Highest growth of the five
Indonesia	not stated	31%	3.9%	Largest economy (PPP)

RESERVE CURRENCY — THE ONE STRUCTURAL DIVIDE

A reserve currency is fully convertible and held by foreign central banks and other investors as part of their FX reserves. When foreigners will hold your currency, you can borrow abroad in your own money — which greatly expands market access, sustains structural deficits and a higher debt level, and minimises the likelihood of default. Emerging and frontier sovereigns with capital controls or limited convertibility have no such option and turn to the IMF and other supranationals instead.

¹ Ratings stated by the curriculum: Malaysia A3 (S&P A), Thailand Baa1, Vietnam Ba3 (S&P BB+). The Philippines matches Thailand; Indonesia is not given.

² Reserve ratio = FX reserves ÷ external debt. Reserves alone would rank Thailand first on external creditworthiness.

Source: CFA Institute, Curriculum Vol 6, Learning Module 15, Examples 6 and 7, printed pp. 390–393. Reserve data Bloomberg, FYE 2020.

SECTION 03

Corporate credit analysis

Traditional credit analysis uses eight Cs — five bottom-up factors specific to the borrower, and three top-down factors that apply to every borrower

Capacity and capital are quantitative — they come off the financial statements. Collateral, covenants and character are qualitative and require analyst judgement.

CONDITIONS

The general economic, competitive and business environment faced by all borrowers, and whether it lets them service or refinance debt.

COUNTRY

The geopolitical environment and the legal and political system faced by all issuers in a jurisdiction that may affect debt payment.

CURRENCY

Affects issuers whose cash flows move with exchange rates, or who borrow outside their own jurisdiction — including sovereigns with FX debt.

CAPACITY

Ability to make debt payments on time.

CAPITAL

Other resources that reduce reliance on debt.

COLLATERAL

Quality and value of assets supporting the debt.

COVENANTS

Legal terms the issuer must comply with.

CHARACTER

Management quality and willingness to repay.

TOP-DOWN (navy, 3) AND BOTTOM-UP (orange, 5)

Top-down Cs describe the environment every borrower in a jurisdiction faces. Bottom-up Cs describe this borrower. Both feed POD; collateral and covenants also move LGD.

Exhibit 3 names four ratios — EBIT margin, EBIT-to-interest, Debt-to-EBITDA and RCF-to-net-debt — and the variants around them are where students lose marks

Leverage can be written with debt on top or on the bottom, so check the direction before you rank: a higher Debt/EBITDA is worse, a higher RCF/net debt is better.

Family	Ratio	Definition	Higher ⇒
Profitability	EBIT margin ★	Operating income ÷ Revenue	Better
Coverage	EBIT / Interest ★	Operating income ÷ Interest expense	Better
Coverage (variant)	EBITDA / Interest	Adds back D&A to the numerator	Better
Coverage (variant)	EBITDAR / (Int+Rent)	Adds rent to numerator and denom.	Better
Leverage (earnings)	Debt / EBITDA ★	Total debt ÷ EBITDA	Worse
Leverage (balance sheet)	Debt / Capital	Debt ÷ (Debt + Equity)	Worse
Leverage (cash flow)	RCF / Net debt ★	RCF ÷ (Debt – cash & mkt sec)	Better

THE CASH-FLOW DEFINITIONS — READ THE FOOTNOTE, NOT THE ACRONYM

FFO = net income from continuing operations + D&A + deferred income taxes + other non-cash items (Question Set, printed p. 434). RCF = FFO – dividends paid; the curriculum also defines it as net cash flow from operating activities less dividends, and in the Bowstream model FFO IS net cash from operating activities. FCF = net income – working-capital increases – investment in fixed assets – net interest paid.

¹ Ratios marked ★ are the four named in Exhibit 3. All are non-IFRS measures — state your definition before comparing issuers.

Source: CFA Institute, Curriculum Vol 6, Learning Module 16, Exhibit 3 and Question Set, printed pp. 418, 434.

A five-year downside walk shows the four ratios deteriorating together — margin compression feeds straight through to coverage and leverage

Assumptions: sales growth 10% then 5%; COGS/sales 65% rising to 85%; interest rate on debt 6% rising to 8%; debt growing 10% a year throughout.

Metric	Y0 (A)	Y1	Y2	Y3	Y4	Y5
EBIT	1,330	1,122	890	632	346	364
EBITDA	1,730	1,589	1,407	1,180	906	916
EBIT margin	25%	20%	15%	10%	5%	5%
EBIT to interest expense	51.9	32.9	23.0	14.4	5.8	5.2
Debt to EBITDA	0.36	0.43	0.54	0.71	1.06	1.21
RCF to net debt	379%	271%	191%	130%	67%	54%

WHAT THE CURRICULUM CONCLUDES — AND WHAT IT DOES NOT

It concludes that Bowstream faces declining profitability, weaker debt coverage and increased leverage, which is likely to lead to wider credit spreads and a potential ratings downgrade. It does NOT assign a rating to any year. The comparison charts that follow (printed pp. 422–423) plot these four ratios against Moody's consumer-durables categories over time — a peer benchmark, not a threshold table. Note that Bowstream stays profitable at EBIT level throughout: it is net income that goes negative in Y4 and Y5, not operating profit.

Source: CFA Institute, Curriculum Vol 6, Learning Module 16, Example 4 (Bowstream Corporation), printed pp. 419–421.

Seniority sets LGD — seven ranks, secured above unsecured — and the realised recovery gap between them is wide and cyclical

All creditors at the same seniority rank as one class: a 30-year senior unsecured bond has exactly the same pro-rata claim in bankruptcy as one maturing in six months.

Rank	Seniority ranking (Exhibit 4)	LGD
1	First lien / first mortgage	Lower
2	Senior secured	Lower
3	Junior secured	Lower
4	Senior unsecured	Higher
5	Senior subordinated	Higher
6	Subordinated	Higher
7	Junior subordinated	Highest

Priority position	2020	2019	1987-2020
Term loans	48.5	58.1	72.6
Senior secured bonds	34.8	45.9	61.4
Senior unsecured bonds	8.6	31.3	46.9
Subordinated bonds	0.9	24.7	27.9

RECOVERY RATES ARE AVERAGES

They vary widely by industry, and they fall when the economy weakens because collateral resale markets thin out. Across industries, first-lien recoveries average 53.0%, senior unsecured 41.8%, subordinated 34.7%.

NOTCHING — THE ISSUER RATING IS THE SENIOR UNSECURED RATING

POD is the same across an issuer's bonds because of cross-default; issue ratings differ because LGD differs. The higher the senior unsecured rating, the smaller the notching adjustment. Curriculum examples: Royal Caribbean carried a B1 corporate family rating with its unsecured issue at B2 and its senior secured notes three notches higher at Ba2; Mojofon's S&P issuer rating of B+ implies a subordinated issue rating of B-.

¹ Term loans include first-lien, second-lien and unsecured. Emergence-year recoveries, North American non-financials.

Source: CFA Institute, Curriculum Vol 6, Learning Module 16, Exhibits 4 and 5, printed pp. 425–427. Recovery data: Moody's Investors Service, Annual Default Study, 28 January 2021.

Losses climb the ladder from the bottom — equity absorbs first, then subordinated, and only what is left over reaches senior unsecured holders

Pari passu within a class, strict priority across classes — and collateral is applied to its own claim first, before anything reaches the general pool.

Claim (most senior first)	Face	Recovery
Secured bank loan	30	30 (100%)
Unsecured bonds	90	80 (88.9%)
Subordinated bonds	10	0 (0%)
Common equity + reserves	70	0 (0%)
Total claims	200	Impairment 90

MOJOJON — WHEN COLLATERAL COVERS THE CLAIM

Capital structure: senior secured bonds 50, pledged against collateral valued at 60; senior unsecured bonds 35; subordinated bonds 15; common equity 100. All bonds carry cross-default.

Because the collateral (60) exceeds the secured bonds outstanding (50), the secured bonds are likely to recover in full. Had collateral been worth less than the claim, the shortfall would rank pari passu with senior unsecured.

CFA LM 16 QUESTION SET Q1 — A 90 IMPAIRMENT

Assets 200: cash 40, net current assets 100, net fixed assets 60. The net current assets are impaired by 90.

1. Equity is most junior: absorbs 70 (shares 40 + reserves 30). 20 left.
2. Subordinated bonds absorb next: total only 10, so recovery is zero. 10 left.
3. The last 10 hits unsecured bonds: $90 - 10 = 80$ recovered, 88.9%.
4. Bank loan untouched: cash collateral of 40 fully covers the 30 claim.

Secured 100% • Unsec 88.9% • Sub 0% • Equity 0%

Source: CFA Institute, Curriculum Vol 6, Learning Module 16, Question Set Q1 (printed p. 431) and Practice Problem 7 (Mojofon Holdings, printed p. 435).

Holding-company debt is structurally subordinated — subsidiary creditors are paid before any cash can be upstreamed, even when both bonds say senior unsecured

Structural subordination is a third rating factor alongside POD and LGD. It is created by where the debt sits in the group, not by what the indenture calls it.

THE MECHANISM

A holding company owns the shares of an operating subsidiary. It does not own the subsidiary's assets.

Operating-subsidary debt is serviced by the cash flow and assets of the subsidiary FIRST. Only surplus cash can be passed — "upstreamed" — to the holding company to service parent-level debt.

In default, the holding company's creditors stand behind the subsidiary's creditors, because what the holding company owns is an equity claim, and equity ranks last.

Result: HoldCo senior unsecured is structurally subordinated to OpCo senior unsecured, even where both are labelled *pari passu*.

THE CURRICULUM'S CASE — PRACTICE PROBLEM 8

Half of the group's senior unsecured bonds are issued by the major operating subsidiary; the other half by the holding company, which relies on dividends upstreamed from subsidiaries.

The subsidiary contributes over 90% of group cash flow and assets. There is no cross-guarantee between holding company and subsidiary.

Answer: the bonds issued by the OPERATING SUBSIDIARY most likely have the higher recovery rate — the holding company's debt can only be serviced once the subsidiary's debt is paid in full.

The three questions to ask any group issuer: what share of debt sits at OpCo versus HoldCo; is there a cross-guarantee; and are there dividend-restriction covenants at OpCo?

Source: CFA Institute, Curriculum Vol 6, Learning Module 16, printed p. 430, and Practice Problem 8 with solution, printed pp. 435–436.

SECTION 04

Securitization structures

Securitization sells an asset pool to a bankruptcy-remote SPE that issues securities against it — without that true sale there is no securitization, only a secured bond

Curriculum example: BRWA sells EUR 1,000 million of car loans (45,000 loans, average balance EUR 22,222) to Car Loan Trust, a legally independent, bankruptcy-remote SPE, and receives cash.

1 SELLER OF THE COLLATERAL (the depositor)

Originates or owns the assets and sells the pool to the SPE. Motive: remove assets and lending risk from the balance sheet, operate more efficiently on a risk-adjusted basis, earn fee income, and expand lending origination beyond the balance sheet.

2 THE SPE (the issuer in the prospectus)

Buys the pool, then issues and sells the ABS. Bankruptcy-remote: if the seller fails, its creditors have no claim on these assets, and courts in most jurisdictions have no discretion to change seniority.

3 THE SERVICER

Administers the loans: collects payments, chases delinquents, repossesses and disposes of collateral. Often, but not always, the originator continues to service.

THIRD PARTIES

Accountants, lawyers, trustees, underwriters, rating agencies and sometimes financial guarantors. The trustee safeguards the assets, holds funds due to ABS holders and issues the periodic reports — the only information investors get.

THE TWO LEGAL DOCUMENTS

The purchase agreement between seller and SPE sets out the representations and warranties about the assets sold. The prospectus describes the structure: priority and amount of payments, and the credit enhancements used.

WHY TRUE SALE MATTERS

All rights are irrevocably transferred and the assets are de-recognised from the seller. The only credit risk investors face is that the underlying borrowers default. Legal frameworks vary by country — and a transfer can still be unwound as a fraudulent conveyance.

Source: CFA Institute, Curriculum Vol 6, Learning Module 17, Exhibits 5–9 and The Role of the SPE, printed pp. 449–453.

Credit tranching stacks bond classes on one pool – the senior class is cheap because the junior classes stand in front of it, and the residual keeps what is left

Same cash-flow pool, three different risk profiles. Tranching dictates both the order of payments and the order in which losses are absorbed.

Bond class	Size (\$)	Rating	Coupon	Attach	Detach	Role in the structure
Senior	70	Aaa / AAA	MRR + 40	30%	100%	Cheapest funding
Mezzanine	20	Baa / BBB	MRR + 250	10%	30%	Yield pick-up
Equity / residual	10	Unrated	Residual CF	0%	10%	First loss

WHERE THE RESIDUAL COMES FROM – POOL OF \$100 YIELDING MRR + 250

The asset pool pays 250 bps over MRR on \$100. The senior \$70 costs MRR + 40; the mezzanine \$20 costs MRR + 250; the equity \$10 takes what is left.

1. Pool spread income: $250 \text{ bps} \times 100 = 25,000 \text{ bps} \cdot \$$
2. Debt cost: $(40 \times 70) + (250 \times 20) = 2,800 + 5,000 = 7,800 \text{ bps} \cdot \$$
3. Residual: $25,000 - 7,800 = 17,200 \text{ bps} \cdot \$$ on equity of 10

Equity excess spread $\approx 1,720$ bps (17.2% p.a.) before any losses, fees or defaults.

Losses are absorbed strictly from the bottom up – the senior class takes nothing until every junior class beneath it has been wiped out

Car Loan Trust: four classes on a EUR 1,000 million pool. Class D absorbs the first 25, then C up to a further 50, then B up to a further 100 — so Class A loses nothing until losses exceed EUR 175 million.

Car Loan Trust bond class	Par (EUR m)	Absorbs losses	Loss at EUR 70m	Repaid
Class A — senior	825	above 175	0	825
Class B — mezzanine	100	75 – 175	0	100
Class C — junior	50	25 – 75	45	5
Class D — most junior	25	first 25	25	0
Total pool	1,000	–	70	930

Ahbaling Trust bond class	Par (SGD m)	Value after a SGD 140m default	Loss
Class A — senior	280	260	20
Class B — subordinated	60	0	60
Class C — subordinated	60	0	60
Total	400	260	140

Source: CFA Institute, Curriculum Vol 6, Learning Module 18, Car Loan Trust structure (printed pp. 470–472) and Ahbaling Trust question set (printed p. 473).

There are exactly three internal credit enhancements — overcollateralization, excess spread and subordination — and everything else is external

The dividing line is whether the support comes from inside the deal or from a third party's promise. Cash collateral accounts sit on the EXTERNAL side — a point the curriculum makes explicitly.

INTERNAL — THE THREE

1. OVERCOLLATERALIZATION — the collateral value exceeds the face value of the bonds issued. Car Loan Trust: EUR 1,200m of loans behind a EUR 1,000m issue. If 10% of the pool defaults there is still EUR 1,080m of collateral.

2. EXCESS SPREAD — the difference between the coupon earned on the collateral and the coupon paid on the securities. If the pool yields 8% and the bonds pay 6%, that 2% absorbs shortfalls or builds reserves.

3. SUBORDINATION / CREDIT TRANCHING — more than one bond class, with the classes differing in how they share losses from defaults in the pool. The tranching sets both the order of payments and the order of loss absorption.

EXTERNAL — A THIRD PARTY'S PROMISE

Financial guarantees by banks or insurance companies.

Letters of credit — a bank commits to advance shortfalls.

Cash collateral accounts — the curriculum classifies these as EXTERNAL, not internal. Do not put a reserve account in the internal list.

The trade-off: external support converts pool risk into counterparty risk. If the guarantor weakens in the same stress that hurts the pool, the enhancement fails exactly when it is needed.

Worked case — SolarCity 2013-1: USD 54.425m of notes against collateral carried at USD 143.9m, roughly 2.7 times cover. That overcollateralization is cited as a main reason the notes achieved BBB+ and priced at 4.8%.

Source: CFA Institute, Curriculum Vol 6, Learning Module 18, Credit Enhancement and Question Set Q4, printed pp. 469, 474, and Self-Assessment Q6, printed p. 463.

Non-mortgage ABS splits on whether the collateral amortizes — credit-card receivables do not, so the structure needs a revolving period and a rapid-amortization trigger

Commercial Finance Partners nine-year credit-card ABS: CHF 900 million of notes against CHF 925 million of receivables, three-year revolving period then six years of amortization.

Tranche	Face (EUR m)	Rate	Credit enhancement feature
Tranche A	400	7.00%	Subordination of Classes B, C and D
Tranche B	300	8.00%	Subordination of Classes C and D
Tranche C	150	10.50%	Subordination of Class D
Tranche D	50	Variable	None – first loss
Notes issued	900	–	Plus CHF 25m overcollateralization

AMORTIZING vs NOT

Auto loans and residential mortgages amortize: each payment includes principal, the pool shrinks, and investors face prepayment and default risk from day one. Credit-card receivables do not amortize — during the revolving period principal repaid is reinvested in new receivables and holders receive only finance charges and fees.

RAPID AMORTIZATION

A trigger that forces early principal amortization if pool performance deteriorates so that repayments no longer replenish the pool. It protects investors — they are repaid earlier and can reinvest elsewhere — but it shortens the life of the bond without warning.

Loss cushion here: CHF 25m of OC plus the CHF 50m Class D = CHF 75m, or 8.33%, before Class C is touched.

SOLAR ABS

AI-Shims issues EUR 300m of notes secured on 10,000 home-improvement loans with a face value of EUR 320m, average balance EUR 32,500, weighted-average rate 7.5%, weighted-average maturity 17 years, all amortizing. Many solar ABS carry a pre-funding period allowing the trust to buy further qualifying assets after closing.

Source: CFA Institute, Curriculum Vol 6, Learning Module 18, Exhibits 5–6, Example 4 and the solar ABS lesson, printed pp. 475–479.

A CLO is a leveraged transaction on a managed pool of senior secured loans — and the overcollateralization test is the covenant that turns equity cash off

CDOs backed by leveraged bank loans are CLOs, by corporate and EM bonds are CBOs, by other CDOs are structured finance CDOs, and by credit default swaps are synthetic. The CLO is the prevailing structure.

CLO tranche	Rated	Return
AAA	Yes	Lowest yield
AA / A / BBB	Yes	Rising yield
BB	Yes	High yield
Equity	No	Residual only

THE COLLATERAL POOL

A diversified portfolio of senior secured loans, generally 100–225 issuers, actively managed by a collateral manager whose skill in selecting and exchanging assets directly drives performance.

The pool is not final at closing: most loans are bought before it closes, then a ramp-up period adds the rest. The manager may replace loans provided they meet the selection criteria.

Flavours: cash flow CLOs (the most common), market value CLOs, synthetic CLOs built with credit derivatives.

THE OVERCOLLATERALIZATION RATIO

A CLO promises USD 700 million of principal to its debt tranches. Investors and rating agencies require the manager to buy more collateral than that.

1. Bank loans purchased: USD 840 million
2. Debt principal promised: USD 700 million
3. OC ratio = $840 / 700 = 1.20$

Each debt tranche has its own OC covenant.

WHEN A TEST IS BREACHED

Cash is diverted away from equity and junior debt to buy more collateral or repay the most senior tranche. That deleverages the CLO — it retires the cheapest funding the deal has. Other tests limit industry concentration, single-obligor exposure and the share of CCC-rated debt.

Source: CFA Institute, Curriculum Vol 6, Learning Module 18, Exhibits 7–9 and the CLO lesson, printed pp. 481–484.

Covered bonds are the simplest structure and the only one with dual recourse — the cover pool stays on the bank's balance sheet, so it is not a true securitization

Originated over 250 years ago in Prussia — the German Pfandbrief — and now issued in Europe, Asia and Australia. Covered bonds usually carry lower credit risk and offer lower yields than otherwise similar ABS.

COVERED BOND

Senior debt obligations of a financial institution, backed by a segregated pool of commercial or residential mortgages or public-sector assets. Ships and aircraft pools also exist.

The loans REMAIN on the issuer's balance sheet, ringfenced into a cover pool. Payment comes from the bank, not from the pool's cash flows.

DUAL RECOURSE: first on the ringfenced cover pool, and if that is insufficient, on the unencumbered assets of the issuing institution.

One bond class per cover pool. The pool is dynamic — a third-party asset monitor checks it, and prepaid or non-performing assets must be replaced. Germany requires at least 2% overcollateralization; loans must meet an LTV cut-off (80% in the curriculum term sheet) or be swapped out.

TRUE-SALE ABS • AND THE REDEMPTION REGIMES

ABS: the pool is SOLD to a bankruptcy-remote SPE and removed from the balance sheet. Single recourse — to the pool only. Typically several bond classes. ABS usually pay floating and pass early payments through; covered bonds typically pay fixed and mature on a fixed date.

Redemption regimes align cash flows with the original schedule if the sponsor defaults:

HARD BULLET — miss the schedule and a bond default is triggered, payments accelerated. The quickest trigger.

SOFT BULLET — default and acceleration are delayed to a new final maturity date, usually up to a year later.

CONDITIONAL PASS-THROUGH — the bond converts into a pass-through security after the original maturity date if payments have not all been made.

Source: CFA Institute, Curriculum Vol 6, Learning Module 18, Exhibits 1–2, Exhibit 7 and Question Set, printed pp. 464–468, 481.

SECTION 05

Mortgage-backed securities

MBS divide on two axes — residential versus commercial, and agency versus non-agency — and only the agency guarantee removes credit risk

In the US, Canada, Japan and South Korea, securities backed by residential mortgages split three ways: guaranteed by a federal agency, guaranteed by a GSE, or issued by private entities with no guarantee at all.

	Agency — federal agency or GSE	Non-agency — private label
RMBS	Federal agency: full faith and credit of the government. GSE: its own guarantee, for a fee.	Private issuers, no guarantee. Credit risk handled inside the structure — pool insurance, subordination.
CMBS	Limited. Pools of commercial mortgages on income-producing property.	The main CMBS market: offices, retail, industrial, multifamily, hotels, health care.

AGENCY RMBS

The guarantee removes credit risk from the investor. What is left is prepayment risk and interest-rate risk — which is why agency MBS analysis is almost entirely about prepayment behaviour.

NON-AGENCY RMBS

Investor bears credit risk as well. Collateral is non-conforming or subprime. Underwriting is read through LTV (loan ÷ property value) and DTI (monthly debt payment ÷ monthly pre-tax income). The sector all but disappeared after 2008.

RECOURSE MATTERS

In a recourse loan the lender can pursue the borrower for any shortfall after selling the property; in a non-recourse loan only the property. US residential mortgages are non-recourse in many states; most of Europe is recourse — which is why strategic default is a US phenomenon.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, Agency and Non-Agency RMBS and Exhibit 8, printed pp. 504, 519.

A heterogeneous pool is summarised by WAC and WAM, both weighted on CURRENT balances — and the pass-through rate is the WAC net of servicing fees

The mortgages in a pool differ in balance, rate and remaining term. WAC and WAM collapse that into two numbers the investor can price.

$$\text{WAC} = \sum w_i \times c_i$$

$$\text{WAM} = \sum w_i \times m_i$$

$$w_i = \text{CB}_i / \sum \text{CB}$$

where: c_i = mortgage rate; m_i = remaining months to maturity; CB = CURRENT balance, not the original. Pass-through rate = WAC – servicing and guarantee fees; the curriculum's worked case is WAC 4.55% – fees 0.55% = 4.00%.

Mortgage	Rate	Beginning balance	Current balance	Orig. term (m)	Months to mat.
A	2.50%	EUR 300,000	EUR 238,000	240	180
B	3.30%	EUR 420,000	EUR 380,000	600	480
C	2.80%	EUR 100,000	EUR 87,000	288	240
D	4.00%	EUR 280,000	EUR 132,000	360	120
E	3.70%	EUR 350,000	EUR 312,000	384	312
TOTAL	–	EUR 1,450,000	EUR 1,149,000	WAC 3.29%	WAM 313 mo.

THE WEIGHTING TRAP

Weight on CURRENT balances (1,149,000), never on beginning balances (1,450,000) — the curriculum's own distractor.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, Exhibits 2 and 4, printed pp. 508–509.

Prepayment risk has two opposite faces — contraction risk when rates FALL and borrowers refinance, extension risk when rates RISE and they stop

The mortgage investor is short the borrower's prepayment option. It gets exercised precisely when the investor would rather it were not.

CONTRACTION RISK — RATES FALL

Definition: the borrower repays principal in a SHORTER time than the contractual schedule, reducing the future payments the investor receives.

Trigger: mortgage rates decline, so homeowners refinance into cheaper loans. Actual prepayments run higher than forecast and the maturity of the MBS is shorter than anticipated at purchase.

Two adverse consequences for the investor:

- principal must be reinvested at LOWER rates;
- the prepayment option reduces potential price appreciation — the bond cannot rally like an option-free bond.

Refinancing is especially common in the US, where prepayment penalties are rare; they are far more common in Europe.

EXTENSION RISK — RATES RISE

Definition: the borrower repays principal over a LONGER period than the contractual schedule, extending the time of repayment and the maturity of the bond.

Trigger: mortgage rates rise, so homeowners keep their below-market loans and may delay new purchases. Actual prepayments run lower than forecast.

The adverse consequence: higher rates reduce the value of the cash flows, those cash flows are now discounted at a higher rate, AND the extension stretches them further out. Both effects push the price the same way.

Balloon risk on a commercial mortgage is a type of extension risk — the lender extends the loan into a workout period rather than being repaid.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, Prepayment Risk, printed p. 498, and Self-Assessment Q5, printed p. 496.

Time tranching cannot eliminate prepayment risk, only redistribute it – the short tranche buys protection from extension, the long tranches from contraction

Sequential pay: all principal goes to Tranche A until it is retired, then to B, then to C. Credit tranching shares LOSSES; time tranching shares TIMING. A deal can have both.

Tranche	Par (USD m)	Coupon	Principal window	Protected against	Curriculum buyer
Tranche A	50	3.75%	Years 0 – 5	Extension risk	Mutual fund
Tranche B	30	3.95%	Years 6 – 15	Contraction risk	Insurance co.
Tranche C	20	4.05%	Years 16 – 30	Contraction risk	Pension fund
Collateral	100	4.00% p/t	WAM 360 mo.	WAC 4.55%	–

WHO PROTECTS WHOM – GET THE DIRECTION RIGHT

Prioritising principal repayment protects the SHORT tranche (A) against EXTENSION risk, and that protection is supplied by the longer tranches B and C. In exchange, B and C are protected against CONTRACTION risk by A, which stands in front of them and absorbs the early principal.

So an investor worried about extension buys Tranche A; an investor worried about contraction buys B or C. The curriculum matches a mutual fund seeking safe returns to A, an insurer with liabilities starting in fifteen years to B, and a pension fund willing to take more risk for higher returns to C. A foundation needing money within a few years and wanting to avoid extension risk also buys A.

Across the range of CMO structures, the more senior a tranche is, the less exposure it has to prepayment risk AND to default risk.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, Exhibit 6 and Knowledge Check, printed pp. 510–512, and Practice Problem 2, printed pp. 514–515.

Five further CMO structures split the pool's cash flows differently — by timing, by principal versus interest, or by leaving one class to collect the residual

Z-tranches and residual tranches are created by TIME TRANCHING. POs, IOs and floaters are created by splitting up the pool's cash flows in other ways.

- 1 Z-TRANCHE (accretion / accrual bond)**
Pays no interest until a pre-set date; during the accrual period the principal value is credited by the stated coupon. Usually the last tranche in a sequential or PAC series. It frees up cash for the other tranches, and its holder faces no reinvestment risk when yields fall. Average lives usually exceed 20 years — risky and hard to value.
- 2 PRINCIPAL-ONLY (PO)**
Receives only the principal repayments from the pool. Holders get face value through scheduled principal and prepayments, so value is highly sensitive to prepayment and interest rates. When rates fall or prepayments accelerate, the PO rises in value.
- 3 INTEREST-ONLY (IO)**
Receives only the interest payments — the companion of the PO. It has no face or par value. Faster prepayments shrink the cash flows paid to IO holders, which is precisely why investors use IOs to hedge portfolios against interest-rate risk.
- 4 FLOATING-RATE AND INVERSE FLOATERS**
Coupons linked to an index or market reference rate, usually with a cap and a floor. Inverse floaters pay MORE as rates fall. Both are used to hedge interest-rate risk, and both are affected by how rate moves change prepayment speeds.
- 5 RESIDUAL TRANCHE**
Collects whatever cash flow remains once every other tranche has been paid. Suited to investors who can carry and hedge that risk — hedge funds and long-term institutions. Banks typically avoid it because of the capital charge.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, Other CMO Structures, printed pp. 512–513, and Self-Assessment Q4, printed p. 496.

A PAC tranche is promised a fixed principal schedule while prepayments stay inside a range — and the support tranche absorbs every deviation from it

PACs are a further evolution of the sequential-pay CMO. The predictability is not free — it is bought from the support tranche, which gives up all of it.

PLANNED AMORTIZATION CLASS (PAC)

Makes scheduled and fixed principal payments over a predetermined period, provided prepayment levels in the pool stay within a specified maximum and minimum range.

Inside that range, ALL prepayment risk is absorbed by the support tranche and the PAC schedule is honoured exactly.

Outside the range the protection weakens: sustained prepayments above the maximum eventually exhaust the support tranche, and sustained prepayments below the minimum mean the support has no principal left to redirect.

Result: a narrow band of average-life uncertainty, a tighter spread, and a natural home with insurers and pension funds that need cash-flow certainty.

SUPPORT (COMPANION) TRANCHE

Takes the entire variability around the PAC schedule. It is the answer to Practice Problem 5: if prepayments are inside the anticipated range, it is the SUPPORT tranche that protects PAC investors from prepayment risk.

Prepayments faster than expected: the excess principal is routed to the support tranche, so its average life **CONTRACTS**.

Prepayments slower than expected: principal is redirected to keep the PAC on schedule and the support receives nothing until the PAC is current, so its average life **EXTENDS**.

Result: highly variable average life, a wider spread, and a buyer base of specialists who are being paid to take convexity risk.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, Other CMO Structures, printed p. 513, and Practice Problem 5, printed p. 524.

The market quotes prepayment speed as a PSA percentage — background you will meet on a desk, but it is not in the CFA Level I curriculum

SMM measures prepayment in a single month; CPR annualises it; PSA expresses CPR as a percentage of a benchmark ramp. Weighted average life, which IS in the curriculum, is what these feed into.

$$\text{SMM} = \text{Prepayment} / (\text{Balance} - \text{Scheduled principal})$$
$$\text{CPR} = 1 - (1 - \text{SMM})^{1/2}$$

where: The 100 PSA benchmark ramps CPR from 0.2% in month 1 by 0.2% a month to 6.0% at month 30, then stays flat. A quoted speed of 150 PSA means 1.5 times the benchmark CPR at every point on that ramp.

Pool month	100 PSA CPR	150 PSA CPR	SMM at 150 PSA	250 PSA CPR	SMM at 250 PSA
1	0.20%	0.30%	0.025%	0.50%	0.042%
6	1.20%	1.80%	0.151%	3.00%	0.254%
12	2.40%	3.60%	0.305%	6.00%	0.514%
30 and beyond	6.00%	9.00%	0.783%	15.00%	1.345%

WHAT THE CURRICULUM USES INSTEAD

Weighted average life — the measure of how long an investor can expect to hold the MBS before it is paid off, assuming rates stay put. The contractual maturity of an MBS does not reveal its actual payments and prepayments.

¹ Kept because CMO prices are quoted at a PSA speed. Not examinable at Level I — the curriculum uses weighted average life.

Source: MARKET CONVENTION — not CFA Level I. The PSA benchmark is an industry convention (Bond Market Association, now SIFMA). SMM, CPR and PSA do not appear anywhere in Curriculum Vol 6.

CMBS trade more like corporate bonds than like RMBS — call protection removes contraction risk, and the balloon payment puts extension risk back in

Curriculum deal: four properties worth GBP 123m secure GBP 100m of mortgages — an LTV of 81.3% — with WAC 4.66% and WAM 88 months, funded by four GBP 25m tranches.

Dimension	RMBS	CMBS
Pool	Thousands of small mortgages	One to a few large mortgages
Concentration risk	Low — highly diversified	High — one loan can dominate
Contraction risk	High when rates fall	Low — call protection
Extension risk	High when rates rise	High — balloon risk
Call protection	None — freely prepayable	Lockout, penalty points, defeasance
Rate convention	Fixed or floating	US fixed; Europe floating (capped)
Key underwriting	LTV and DTI	LTV and DSC = NOI / debt service

BALLOON RISK IS EXTENSION RISK

Commercial loans are not self-amortizing: they pay interest and some principal, then repay the unamortized balance as a large balloon at year 5 or 10. Balloon risk is the risk the borrower cannot make that payment — because refinancing is unavailable on deteriorating DSC or LTV, because the lender will not extend, or because the property cannot be sold. The lender then extends the loan into a workout period and modifies its terms, which is why the curriculum classifies balloon risk as a TYPE of extension risk. $DSC = NOI \div \text{debt service}$, where NOI is rental income less cash operating expenses less a replacement reserve; a DSC above 1.0x means the property covers its debt service while staying in its initial state of repair.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, CMBS Structure, Exhibits 7–8 and Practice Problem 4, printed pp. 516–521.

Deck 16 — Fixed Income IV — Credit Risk, Credit Analysis & Securitization

■ Credit risk = expected loss = $POD \times LGD$, where $LGD = EE \times (1 - RR)$. RR is what you recover; $(1 - RR)$ is severity. A fair credit spread approximates that same product.

■ Sovereign creditworthiness needs ability to pay AND willingness to pay, because sovereign immunity blocks enforcement. Reserve-currency status is the one structural divide.

■ Exhibit 3 names four corporate ratios — EBIT margin, EBIT/interest, Debt/EBITDA, RCF/net debt. Ratios drive POD; seniority drives LGD; notching connects issuer to issue rating.

■ Securitization needs a true sale to a bankruptcy-remote SPE. There are exactly three internal credit enhancements: overcollateralization, excess spread and subordination.

■ The investment-grade line sits at Baa3 (Moody's) and BBB- (S&P, Fitch). Credit migration costs money without any default, and the spread moves long before the rating does.

■ The Cs of credit analysis are eight: five bottom-up (capacity, capital, collateral, covenants, character) and three top-down (conditions, country, currency).

■ Losses climb the waterfall from the bottom: equity, then subordinated, then senior unsecured. HoldCo debt is structurally subordinated to OpCo debt whatever the label says.

■ Prepayment risk has two faces: contraction when rates FALL, extension when rates RISE. Credit tranching allocates losses; time tranching allocates timing. Balloon risk is extension risk.